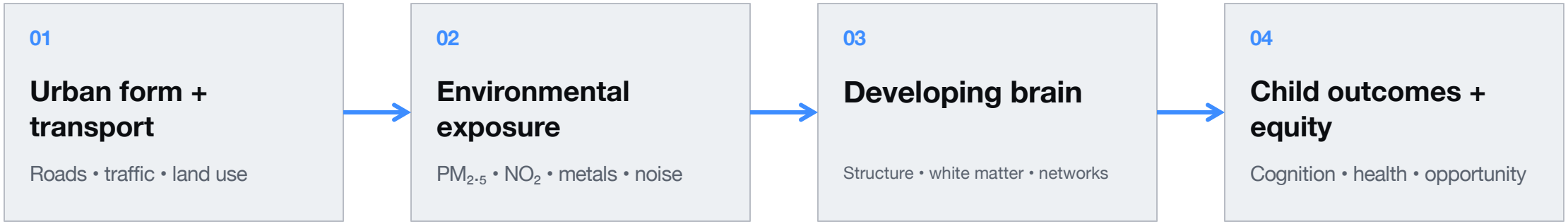
Urban planning →children brain development

How transportation, environmental exposure, and urban inequality may shape the developing brain



The question is urban + biological

How do traffic-related pollution and noise relate to children's brain development?
How should the cities change to benefit children/adolescence brain development?



The city development varies



1. **Roads + traffic**
   Volumes, diesel fleets, braking, commuting patterns

2. **School + housing siting**
   Distance to highways, time spent outdoors

3. **Public area**
   Green space, quiet sides, street design

4. **Unequal urban economic**
   Investment varies between each neighborhood

The housing, schools, roads are strongly connected, depending on the government planning decisions.
Thus, urban development may influence brain development in an early stage

Seven papers in total

7 papers reviewed

3 Data base

1 methods paper

BREATHE · Barcelona	**2 papers**	Same MRI subgroup (n=263)
Generation R · Rotterdam	**1 paper**	Pollution + road-noise study (n=2,197)
ABCD · United States	**3 papers**	Overlapping national cohort
Gordon atlas	**1 paper**	Measurement foundation for ABCD studies

Study 1:

Copper at school

Focus: copper in schoolyard $PM_{2.5}$, linked mainly to brake pads and industrial activity.

Method:

Experimental objects:
2,836 Barcelona children, ages 8–12; MRI subgroup n=263, from 39 schools.

Frequency:
two one-week outdoor sampling campaigns within a year

Testing tools:
ANT motor task + T1 MRI + DTI(Diffusion Tensor Imaging) + resting-state fMRI.

Modeling:
Multiple linear regression + linear regression + voxel-wise SPM regression with FWE correction.

Statistical Modeling Methods

1. Multiple linear regression

The study is trying to explain where the copper might come from, and the x values are elemental carbon(road traffic source), zinc(industrial source), and train proximity(railway source). Y value are copper level.

They found that road traffic> industry > railway in terms of contributing to copper level. (0.66>0.31>0.20)

2. Linear regression

The study wants to find out whether higher copper exposure was associated with poorer motor performance in children.

X: copper concentration at the child's school
Y: ANT motor-task performance, including reaction time and varability.

Results

- +1 ng/m^3 Cu → reaction time +2.2 ms
- Reaction-time variability +2.9 ms
- Higher Cu → altered caudate tissue
- Weaker caudate–frontal connectivity: controlling movement, attention, and information processing.
- Results persisted after counting socioeconomic(lower income neighborhood) conditions and co-pollutant adjustment



Interpretation of Key Findings

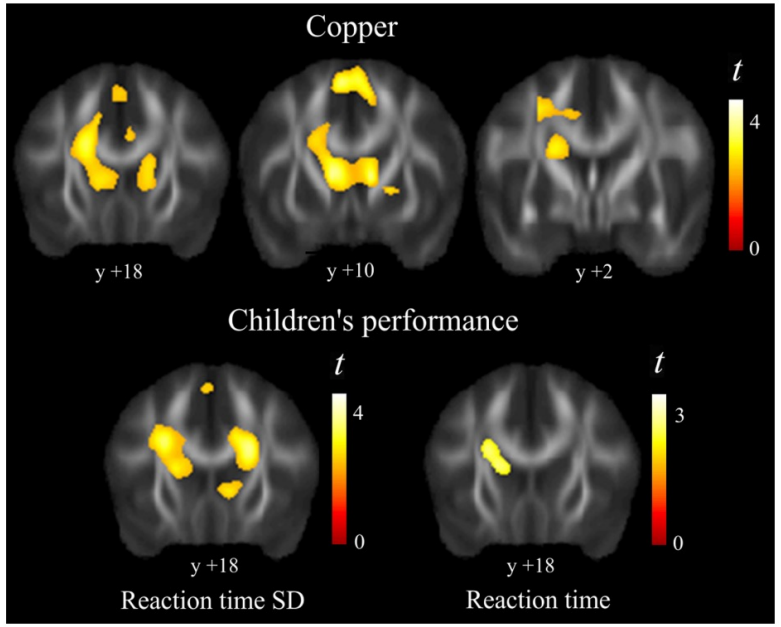


Figure 2. Diffusion tensor
Higher copper levels correl
FA (fractional anisotropy) p
caudate nucleus region (to
correlation with motor per
(bottom panel) showed bo
reaction time and larger re
standard deviation associat
FA in white matter adjacer
nucleus. The right hemisph
to the right side. *Y* denote
coordinates.

- The yellow region represents the brain areas where the relationship is statistically significant.

- In the top panel, the yellow area shows the brain regions where higher copper exposure was associated with higher FA (Fractional Anisotropy). The author indicated as the white-matter fibers around the caudate may be less developed.

- In the lower panel, it shows the reaction time standard deviation and reaction time.

- The higher FA was associated with poorer motor performance, larger reaction time-SD and longer reaction time.

Study 2

Focus:
traffic-related air pollution in school areas, measured mainly by EC and NO_2. These are strongly related to vehicle exhaust, especially diesel traffic.

Method:
Same data as study 1.

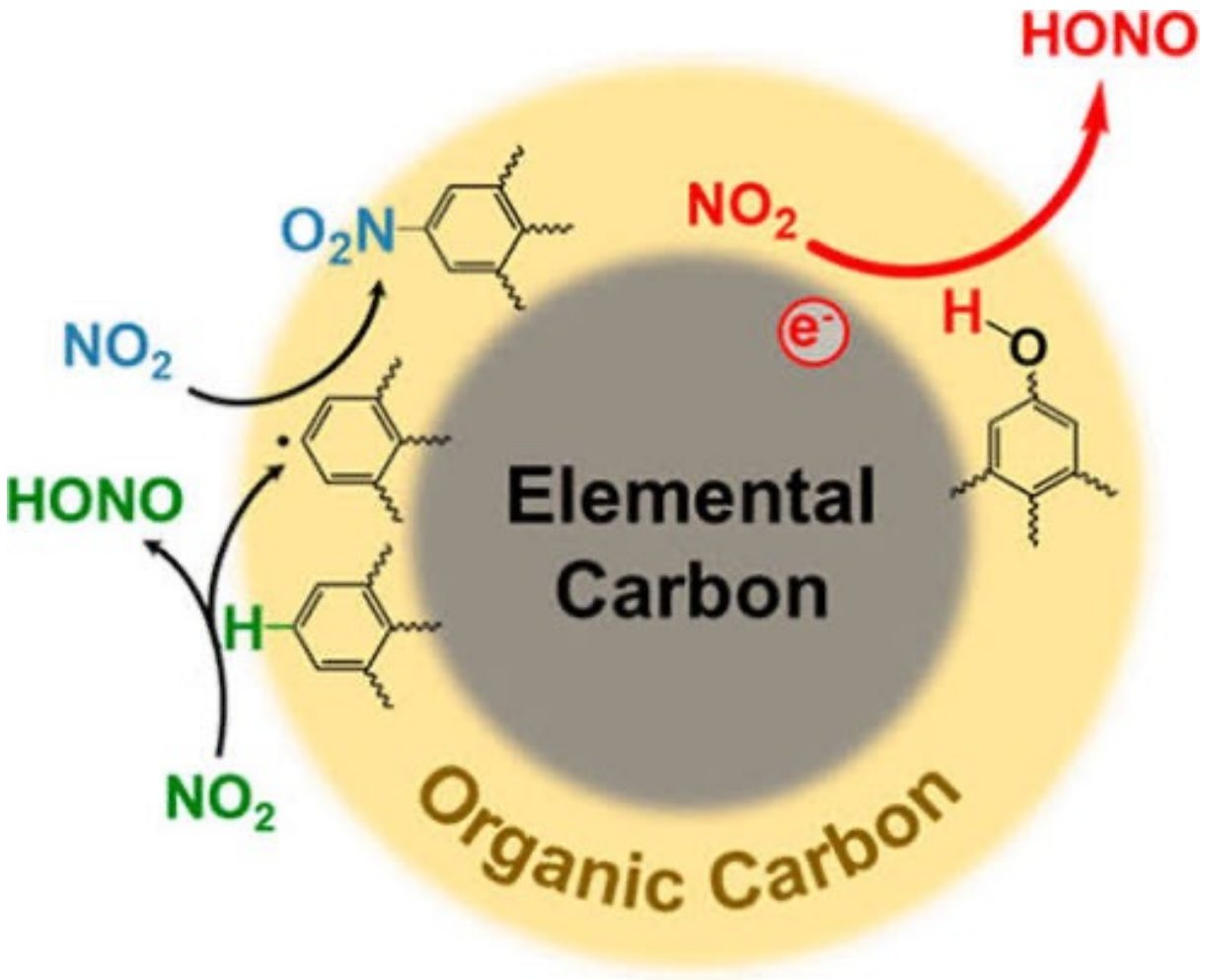
Used EC(Elemental carbon) and NO2 (Nitrogen dioxide) as indicator of pollution.

Other method were the same to study 1, including seed-based resting-state fMRI.

Result:
Higher pollution → weaker connectivity

Higher pollution → slower reaction time

Pollution →less mature brain-connectivity pattern
Grow older→ brain network strongly connect within the same, more separated from other networks



Results:

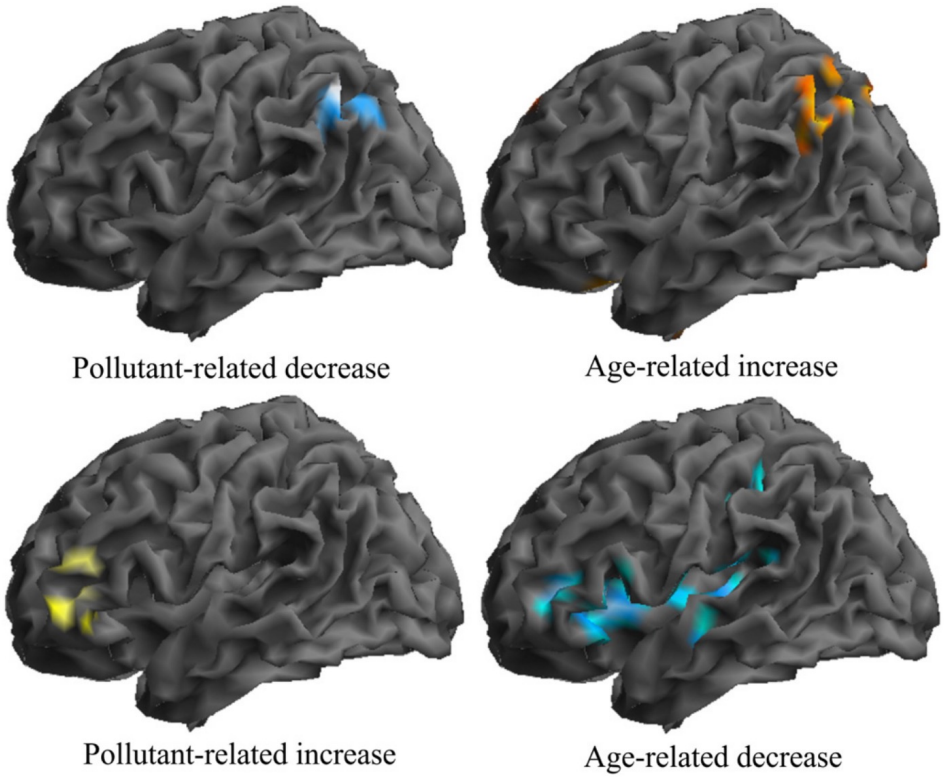


Fig. 3. 3D rendering display of urban pollution and age effects in the medial frontal seed map. Age and air pollution were associated with opposite effects on functional connectivity in notably similar areas, thus indicating that higher exposure may interfere with the normal development of functional connections.

Top left: children with higher pollution exposure tended to have weaker connectivity with these blue regions.

Top right: older children tended to have stronger connectivity with those regions.

Bottom left: Higher pollution was connected to increased connectivity in the yellow regions.

Bottom right: connectivity in these regions decreased with age.

Study 3

Focus: Traffic-related air pollution during pregnancy and childhood → NO_2, NOx, and $PM_{2.5}$ absorbance,

Method: **2,197 children** from the Generation R Study, ages **9–12**

They mainly estimated home address based on air pollution and road-noise

Testing time:

Pregnancy

Birth–3 years

3–6 years

6 years until the MRI scan at age 9–12

Linear models with multiple-testing correction



Models:

- There are 380 brain regions.
- Multiple linear regression, 71,820 total unique connections.
- Doing significant test and False Discovery Rate correction.
- 26 brain regions are significantly related to pollution
- 20/26 connected regions from two different brain networks

Results:

Higher early-childhood air pollution → increased cross-network connectivity → altered brain-network organization
Traffic noise → no significant effect

Significant chord graph

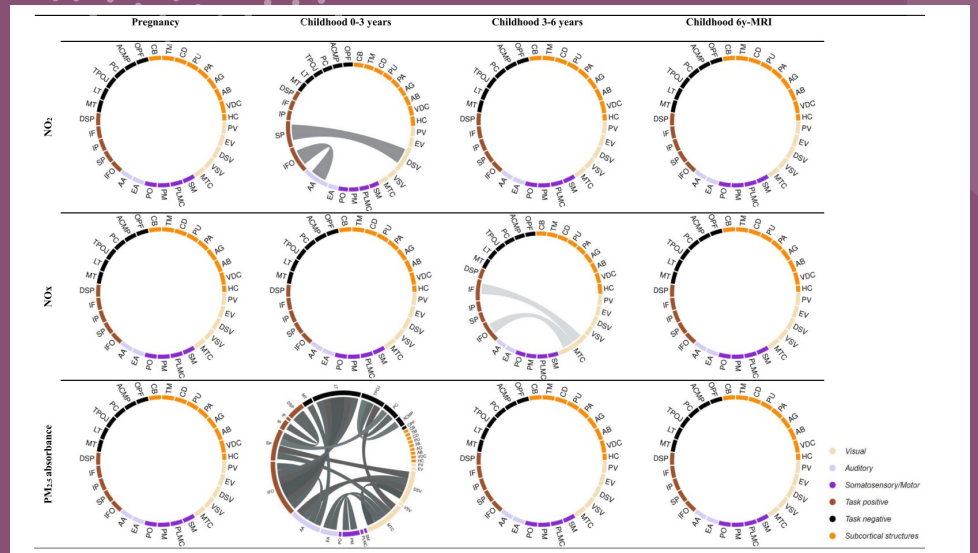
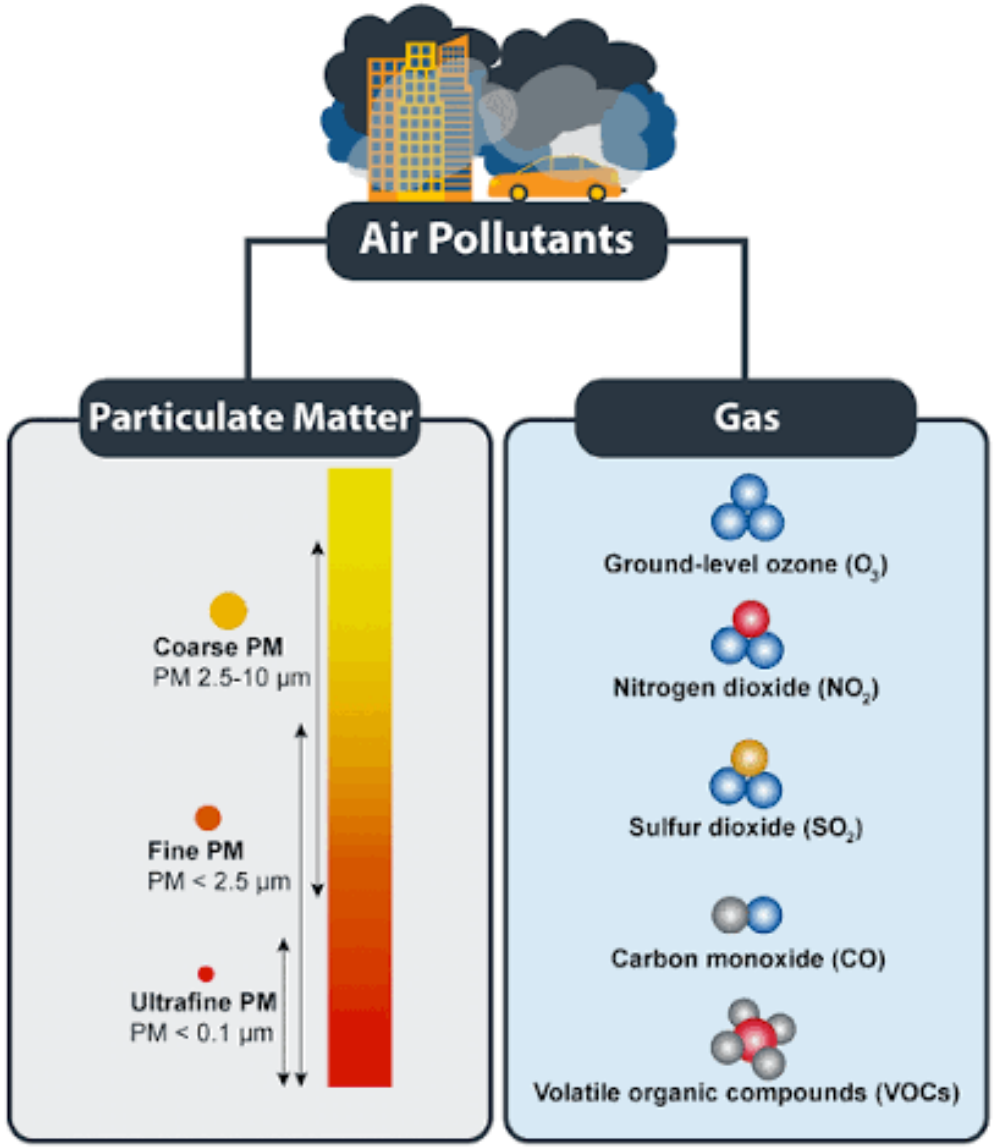


Fig. 2. Adjusted associations between exposure to air pollution at each time period and functional brain connectivity in preadolescents. Brain areas were grouped into 31 brain regions (described below) and into 5 different brain functional networks: visual, auditory, somatosensory/motor, task positive, task negative (also known as Default Mode Network (DMN)), and a 6th group with the subcortical structures and the cerebellum: AA, Auditory Association Cortex; AB, Nucleaus Accumbens; ACMP, Anterior Cingulate and Medial Prefrontal Cortex; AG, Amygdala; CB, Cerebellum; CD, Caudate; DSP, DorsoLateral Prefrontal Cortex; DSV, Dorsal Stream Visual Cortex; EA, Early Auditory Cortex; EV, Early Visual Cortex; HC, Hippocampus; IF, Inferior Frontal Cortex; IFO, Insular and Frontal Opercular Cortex; IP, Inferior Parietal Cortex; LT, Lateral Temporal Cortex; MT, Medial Temporal Cortex; MTC, MT + Complex and Neighboring Visual Areas; NOx, nitrogen oxides; NO2, nitrogen dioxide; OPF, Orbital and Polar Frontal Cortex; PA, Pallidum; PC, Posterior Cingulate Cortex; PLMC, Paracentral Lobular and Mid Cingulate Cortex; PM, Premotor Cortex; PM2.5absorbance, absorbance of PM2.5 filters; PO, Posterior Opercular Cortex; PU, Putamen; PV, Primary Visual Cortex; SM, Somatosensory and Motor Cortex; SP, Superior Parietal Cortex; TPOJ, Temporo-Parieto-Occipital Junction; TM, Thalamus; VDC, Ventral Diencephalon; VSV, Ventral Stream Visual Cortex. Associations from linear regression models adjusted for maternal and paternal education, ethnicity, age, body mass index, and psychological distress during pregnancy, maternal smoking and alcohol use during pregnancy, parity, and intelligence quotient, family status, household income, child's gender and age at the scanning session, and mean framewise displacement that survived correction for multiple testing using false discovery rating. All associations showed positive coefficients and the color of the connection represents the strengths of the association (the darkness of the color indicates a larger beta coefficient). Linear regression models were performed separately for each air pollutant.



Study 4

Focus:

- How $PM_{2.5}$, NO_2, and O_3 exposure affects the development of brain connectivity from Late childhood to early adolescence.

- Pollution→ estimated in children's home addresses

Method:

- 9,497 children from ABCD Study
- Ages 9–10
- Used resting-state fMRI
- Linear mixed-effects models

Results

$PM_{2.5}$:

higher exposure → more connectivity between different cortical networks → reduced normal segregation when growing up

NO_2:

higher exposure → less inter-network connectivity

O_3:

higher exposure →more within-network connectivity, but less subcortical-to-network connectivity→ potentially less flexible communication between brain systems.

Study 5

- Focus: How exposure to $PM_{2.5}$, NO_2, and O_3 may affect brain connectivity through the immune system

- Method: 5088 adolescents(11-13) from the ABCD Study

    Measured blood immune markers

    Used resting-state fMRI

    Used PLSC(Partial Least Squares Correlation) analysis

Results

- Air pollution: higher childhood exposure → differences in white-matter microstructure

- higher childhood exposure → lower white blood cell count in males → suggests potential changes in immune function

- For women no significant relationship.

- Females: higher pollution exposure →higher diffusion measures in the fornix and superior corticostriate tract.

- Males: higher pollution exposure→ lower diffusion measures across several major white-matter tracts→ maybe less developed(Slower information processing, possible impact to memory)

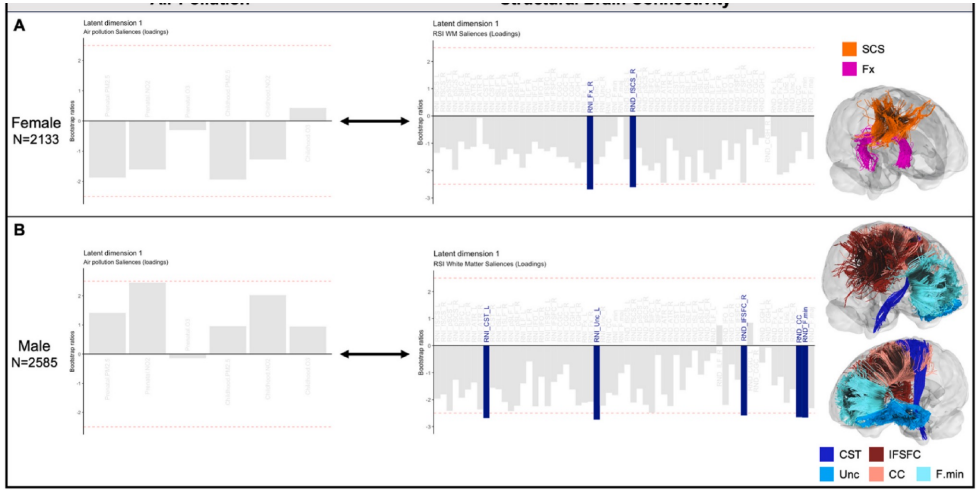


Fig. 1. Bar plots depicting bootstrap ratios of air pollution and structural brain connectivity loadings for the identified latent dimension in female and male adolescents. Abbreviations: restriction spectrum imaging (RSI), restricted normalized isotropic diffusion (RNI), restricted normalized directional diffusion (RND), right hemisphere (R), left hemisphere (L), Fornix (Fx), superior corticostriate (SCS), corticospinal tract (CST), uncinate fasciculus (Unc), inferior frontal to superior frontal cortex (IFSFC), forceps minor (F.min), and corpus callosum (CC).

Gordon studies

- Focus:

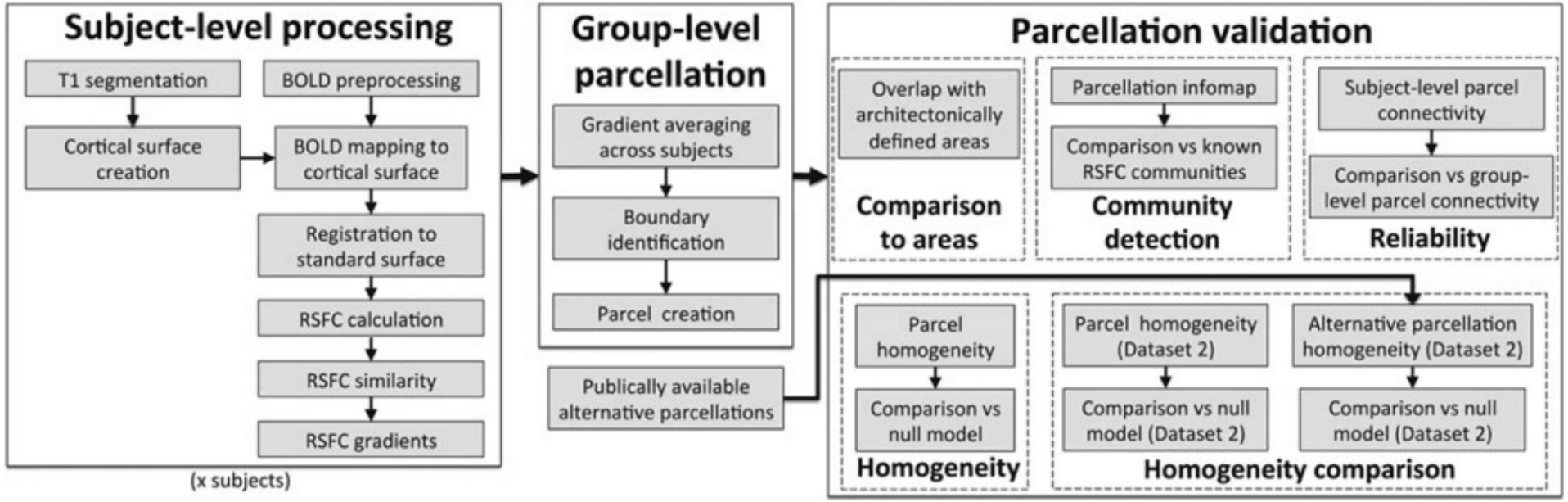
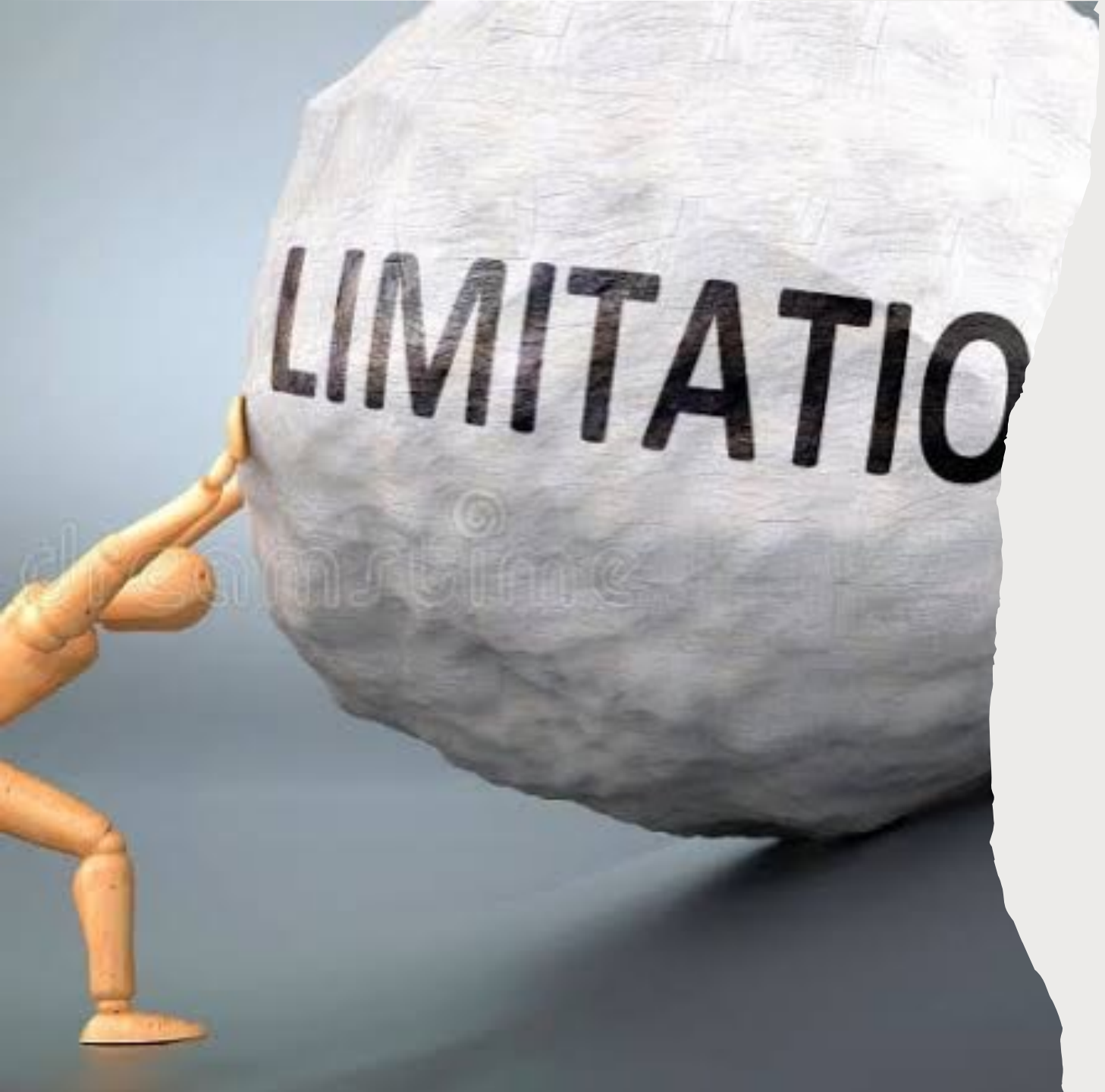


Figure 1. Visual outline of analysis methods.



Limitations

1. Exposure is estimated, not directly measured exactly

Home or school exposure not equal to total daily exposure of the children, confounding variables

Observational, so they show association, no causation

Variable pollutants are correlated, hard to separate and have a clear result

fMRI connectivity is an indirect measure of brain function

Brain-connectivity differences didn't mean a child has a cognitive problem, such as memory

Results differ across studies because they use different pollutants, ages, exposure periods, and connectivity methods

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study 2

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